

LUCID-100 Replication Report — Slot 69

Combined DNA Damage Repair Interference and Ion Beam Therapy:
Development, Benchmark and Clinical Implications of a Mechanistic Biological Model
(Liew, Meister, Mein, Tessonnier, Kopp, Held, Haberer, Abdollahi, Debus, Dokic, Mairani,
IJROBP 112(3):802–817, 2022)

OpenClaw / Ollie subagent (LUCID-replications backfill)

2026-07-06 (backfill from live evidence dated 2026-06-09 through 2026-06-22)

1 Paper summary

The target paper (DOI 10.1016/j.ijrobp.2021.09.048, PMID 34710524, Elsevier closed-access) reports the development and benchmarking of a mechanistic biological effect model — de-masked here as **UNIVERSE** (UNified and VErSatile bio-response Engine, DKFZ/HIT, Liew/Mein/Mairani) — built on the GLOBLE giant-loop binary-lesion framework of Friedrich et al. 2012. UNIVERSE is the *combination* of two previously published open-access extensions:

1. The photon DNA-damage-repair-interference (DDRi) extension (Liew et al. 2019, *IJMS* 20:6054).
2. The ion-beam Kiefer–Chatterjee track-structure extension with Friedrich 2015 intra-track DSB clustering (Mein et al. 2019, *Radiat Oncol* 14:100).

The paper’s headline claims are (i) that UNIVERSE predicts photon and helium ion cell survival with three parameters derived from photon data alone; (ii) that DDR interference produces monotone dose-dependent radiosensitisation that *diminishes with increasing dose and LET*; and (iii) that a helium SOBP measurement campaign in repair-competent vs. repair-deficient cell lines supports the model and, when combined with recalculated patient plans, predicts that DDRi + particle preserves the therapeutic window better than DDRi + photon. The novel contributions are the ion-beam experiment set and the patient-plan translation; the mechanistic framework itself is stated fully in the two OA twin papers.

2 Claims table

#	Claim	Type	Testable?	Tested?
C1	UNIVERSE predicts photon survival of A549, H460, H1437, B16, Renca with 3 parameters.	quantitative	yes (OA equations, published K values)	Yes — verified
C2	DDR interference produces dose-dependent radiosensitisation; RSF multiplies only K_{DSB} .	quantitative	yes (OA Eq. 7 + Table 3 RSFs)	Yes — verified

#	Claim	Type	Testable?	Tested?
C3	DDRi sensitisation <i>diminishes with increasing dose</i> .	quantitative	yes (from the smoke)	Yes — verified
C4	DDRi sensitisation <i>diminishes with increasing LET</i> .	quantitative	shape yes, absolute no (needs full Kiefer–Chatterjee)	Qualitative — shape verified
C5	RBE _{no-DDRi} rises with LET.	quantitative	shape yes	Qualitative — verified
C6	Novel helium-SOBP cell-survival measurements at HIT.	experimental	aho (raw data closed)	No
C7	Same 3 parameters predict protons + ⁴ He across clinical LET range.	structural + quantitative	structural yes; quantitative needs full ion MC	Partial — structural yes
C8	Patient plan recalculation: DDRi + particle preserves therapeutic window vs. DDRi + photon.	clinical translation	no (closed HIT TPS + patient CT/RT)	No

Of 8 identified claims: 5 verified (C1, C2, C3, C4-shape, C5-shape), 1 partial (C7), 2 not tested (C6, C8). Testable-claim coverage = $6/8 = 75\%$; verified/all = $5/8 = 62.5\%$.

3 Method (reproduction)

- Model re-implementation.** Wrote `code/universe.smoke.py` (266 LOC, NumPy + SciPy). Implements: (a) GLOBLE giant-loop nuclear geometry (`DNA_c=6000` Mbp, `DNA_g1=2.0` Mbp, `N_g1≈3000`); (b) per-iteration MC: $N_{\text{tDSB}} \sim \text{Poisson}(\alpha_{\text{DSB}} \cdot D \cdot \text{DNA}_c)$; multinomial(uniform) distribution across loops; per-loop classification $1 \text{ DSB} \rightarrow \text{iDSB}$, $\geq 2 \text{ DSBs} \rightarrow \text{cDSB}$; (c) survival expression $S = (1 - K_{\text{iDSB}})^{N_{\text{iDSB}}} \cdot (1 - K_{\text{cDSB}})^{N_{\text{cDSB}}}$, MC-averaged; (d) DDR interference via RSF on K_{iDSB} (Liew 2019 Eq. 7).
- Ion-LET surrogate.** A single bounded analytical form for $\alpha_{\text{DSB}}(\text{LET})$ anchored to published ⁴He RBE-vs-LET behaviour (Mein 2019). This is the *only* methodological substitution and is documented in `source/model_notes.md` §7. The full Kiefer–Chatterjee radial dose distribution + Friedrich 2015 intra-track clustering was *not* implemented (Friedrich 2015 formula is paywalled; RDD chain is $\gtrsim 300$ LOC).
- Driver.** `code/run.smoke.py` (227 LOC) runs three sweeps:
 - Photon dose response, 5 cell lines, $D \in \{0.5, 1, 2, 3, 4, 5, 6, 8, 10\}$ Gy.

- (b) Photon + ATMi, H460 + H1437, RSF $\in \{1.0, 1.73/1.77, 2.56/2.52, 4.21/3.77\}$ (Liew 2019 Table 3 verbatim).
 - (c) LET sweep at 4 Gy, LET $\in \{2, 5, 10, 20, 30, 50, 80, 120\}$ keV/ μ m, with and without DDRi (RSF=2.5).
4. **Cell-line parameters.** K_{iDSB} and K_{cDSB} taken verbatim from Liew 2019 Table 1 for the 5 lines A549, H460, H1437, B16, Renca.
 5. **Compute.** CherryRd (macOS, single-thread NumPy). End-to-end runtime 26.9 s per full smoke (re-verified during report write). No GPU, no paid endpoints, no closed software.
 6. **Tools/versions used.** Python 3, NumPy, SciPy (via macOS Homebrew stack, versions as of 2026-06-09). Verdict logic + reporting: manual + Argo LLM (localhost:44497, model argo:claude-opus-4.8, key stevens).
 7. **Command to reproduce.** `cd code && python3 run_smoke.py`. Deterministic (fixed MC seed) — regenerates all 5 result CSV/JSON + 3 PNG figures byte-equivalently.

4 Results vs. paper

4.1 Photon LQ fits (C1)

Cell line	α (Gy $^{-1}$)	β (Gy $^{-2}$)	α/β (Gy)	SF@2Gy	SF@6Gy
A549	0.150	0.022	6.77	0.68–0.80 range	0.26–0.37 range
H460	0.103	0.033	3.10	idem	idem
H1437	0.119	0.018	6.65	idem	idem
B16	0.124	0.018	6.93	idem	idem
Renca	0.063	0.027	2.34	idem	idem

All α/β values fall inside published in-vitro ranges for the same lines.

4.2 DDRi dose-response steepening (C2, C3)

For H460 the smoke gives SF@2Gy = 0.716 (DMSO) $\rightarrow$ 0.621 (100 nM ATMi) $\rightarrow$ 0.531 (200 nM) $\rightarrow$ 0.385 (500 nM); SF@6Gy = 0.163 $\rightarrow$ 0.110 $\rightarrow$ 0.069 $\rightarrow$ 0.028 (a $\sim 6\times$ drop at the steepest condition). H1437 mirrors the pattern. The DDRi/no-DDRi SF-ratio at H460/500nM shrinks monotonically with dose: 0.854 (0.5 Gy) $\rightarrow$ 0.734 (1) $\rightarrow$ 0.537 (2) $\rightarrow$ 0.397 (3) $\rightarrow$ 0.296 (4) $\rightarrow$ 0.224 (5) $\rightarrow$ 0.169 (6 Gy) — direct confirmation of the “dose” half of the paper’s “diminishes with dose or LET” claim.

4.3 LET sweep (C4, C5) — headline mechanistic test

LET (keV/ μ m)	RBE _{no-DDRi}	RBE _{DDRi}	Ratio
2	0.998	3.337	3.343
5	0.989	3.343	3.379
10	0.993	3.457	3.481
20	1.019	3.808	3.737
30	1.019	4.003	3.929 (peak)
50	1.130	4.386	3.880
80	1.351	4.579	3.389
120	1.599	4.627	2.895

The RBE-ratio curve is non-monotone: it rises to a peak of 3.93 near LET ≈ 30 keV/ μ m and then falls to 2.89 at 120 keV/ μ m. This shape is exactly the mechanistic prediction of the paper (DDRi loses leverage at high LET because K_{cDSB} is invariant under repair-pathway interference and clustered lesions dominate). RBE_{no-DDRi} rises $\sim 1.0 \rightarrow 1.6$ across the sweep, consistent with published ^4He RBE-LET curves.

5 Per-claim what-worked / what-didn't

C1 (photon LQ) — worked. Full re-implementation from Liew 2019 Eqs. 1–3 + Table 1 K values reproduces sensible LQ fits for all 5 published cell lines. No fitting drama.

C2 (DDRi RSF-on- K_{iDSB}) — worked. Directly implementing Eq. 7 with the Table 3 RSFs reproduces the monotone steepening. This is the strongest piece of evidence because the RSFs are taken verbatim and the observation matches Liew 2019 Fig. 3.

C3 (dose attenuation of DDRi effect) — worked. Emerges naturally from the model: at high dose the cDSB channel (unaffected by RSF) already dominates lethality.

C4 (LET attenuation of DDRi effect) — shape only. The peak-then-fall shape of the RBE-ratio curve is reproduced qualitatively via the LET surrogate. The absolute numbers are surrogate-driven and **are not** a quantitative reproduction of the paper's RBE-vs-LET table.

C5 (RBE rises with LET) — shape only. Monotone rise $1.0 \rightarrow 1.6$ across 2–120 keV/ μ m. Order-of-magnitude and monotonicity match published ^4He curves; absolute numbers not a quantitative reproduction.

C6 (helium SOBP experiment) — not tested. Raw survival data (cell line $\times$ depth $\times$ dose $\times$ SF $\pm$ SD) is not in any OA supplement or repository. Would require an author-data request.

C7 (3-param generalisation to ions) — structurally yes, quantitatively partial. The smoke re-uses only the 2 photon K values + RSF; no new parameters are introduced for the LET sweep. This is the paper's structural claim and it is honoured. The quantitative per-ion RBE(LET) numbers are not reproduced.

C8 (patient plan recalculation) — not tested. Requires the closed HIT FLUKA-coupled TPS + anonymised CT + RT-Plan.

6 Critique of the replication (honest)

This section names the shortcuts, weak evidence, and unverified claims of the *replication itself*, per Rick’s 2026-07-05 rule.

1. **The LET surrogate is a shortcut, not a reproduction.** The single biggest methodological substitution: we did not implement the Kiefer–Chatterjee radial dose distribution, the Barkas effective-charge correction, the radial diffusion convolution, or the Friedrich 2015 intra-track DSB-clustering formula. What we did was pick a bounded analytical $\alpha_{\text{DSB}}(\text{LET})$ curve tuned to reproduce the published RBE(LET) shape. In consequence, C4 and C5 are honestly *shape verifications, not quantitative reproductions*. The RBE-ratio “peaks at 30 keV/ μm at 3.93” number is a *property of our surrogate*, not an independent test of the paper’s number. Any reader who reads C4 as “we quantitatively reproduced the paper’s headline RBE-ratio” is being misled; the correct reading is “we constructed a bounded model whose free parameters were chosen consistently with the published RBE-vs-LET shape, and that model produces a non-monotone RBE-ratio with the topology the paper predicts.”
2. **Absolute-value comparisons vs. the paper are absent.** We do not overlay our RBE(LET) curve on the paper’s Figure 2 or Figure 5 and compute a residual. The paper is Elsevier-closed and we only have the abstract; figure digitisation from the PDF was not attempted. Consequence: we cannot say whether our surrogate agrees with the paper to 10%, 30%, or a factor of 2 in absolute terms.
3. **No uncertainty quantification.** The MC in `universe.smoke.py` runs a fixed number of iterations and reports point estimates. There is no bootstrap of the SF or LQ fits, no confidence interval on the RBE-ratio peak location, no sensitivity analysis on the LET-surrogate parameters. A serious replication would at minimum sweep MC iteration counts to establish convergence and quote CIs.
4. **DDR-deficient lines were skipped.** Liew 2019 Table 3 also gives RSF for CHO V3 (DNA-PKcs $^{-/-}$, RSF ≈ 10) and xrs-5 (Ku80 $^{-/-}$, RSF ≈ 15). Adding them is $\lesssim 10$ LOC. Not doing so is a smoke-completeness shortcut, and it means we have no test of whether the RSF-on- K_{DSB} mechanism breaks down at very high RSF (where survival at 6 Gy becomes numerically negligible and the model’s assumptions may strain).
5. **Ion identity is erased.** The LET sweep is ion-agnostic. The paper explicitly compares protons and ^4He and derives different RBE(LET) curves for each. We do not distinguish them. This is another consequence of using a surrogate instead of the full RDD chain; it also means we cannot test the paper’s implicit claim that the same 3 parameters predict *both* species without species-specific tuning.
6. **No hypoxia arm.** Liew 2020 gives HRF parameterisation ($m = 2.94$, $K = 0.129\%$). It was not exercised. Not a headline of the 2021 IJROBP paper but relevant for clinical interpretation of DDRi + particle.
7. **No patient plan.** Zero coverage of C8. The paper’s clinical-translation claim is untested. Marking this as “not tested” is honest, but the AUDIT_PROTOCOL definition of “replicated” probably requires a comment on whether the clinical result is even *plausible* on its face — we do not attempt that.

8. **Verdict-vs-scope mismatch.** The report is labelled REPLICATED in the priority queue, but the internal REPORT.md verdict is **PARTIAL** with coverage 5/10 and agreement 7/10. AUDIT_PROTOCOL requires $\geq 80\%$ coverage for full replication and this run is $\sim 50\%$. The queue’s REPLICATED label is preserved for this backfill only because the mechanistic-core reproduction is unambiguous; the honest scientific verdict is **PARTIAL / mechanistic core only**. Both labels are recorded here so a downstream rollup can reconcile.
9. **Paper-side triangulation is missing.** We ground everything in the OA twin papers (Liew 2019, Mein 2019, Liew 2022) and the abstract of the 2021 target. We have not read the 2021 paper in full because it is Elsevier-closed and we did not do figure OCR. This limits our critique of the paper: we cannot, for example, say whether the paper adequately controls for cell-line drift between the DDRi experiments and the SOBP experiments.

7 Verdict + justification

The priority-queue verdict is **REPLICATED**. The internal REPORT.md verdict is **PARTIAL** (COVERAGE=5/10, AGREEMENT=7/10). Per Rick’s backfill rule, the original verdict is preserved (REPLICATED); the honest, more conservative verdict from the actual evidence is **PARTIAL** (mechanistic core only). The two views reconcile as: the mechanistic model *is* independently reproduced from open sources; the novel experimental and clinical contributions *are not* reproduced and are not reproducible on CherryRd without author-data + closed HIT software. Downstream users should treat this record as “PARTIAL / mechanistic core” unless they have a specific reason to accept the more generous label.

8 Open Questions

The following five questions are grounded in a re-read of the paper (via OA twins, target abstract, and this replication’s own gaps) and are intentionally chosen to be *truly open* — each with concrete next steps.

Q1. *Does the DDRi effect really diminish with dose and LET simultaneously in experimental cell-survival data, or does that co-attenuation live only inside the UNIVERSE ansatz?*

Basis. UNIVERSE assumes RSF multiplies only K_{IDSB} and that K_{cDSB} is invariant under repair-pathway interference. Both attenuations (with dose *and* with LET) fall out of that single algebraic choice; they are not independent observations. If the invariance assumption is wrong, the twin attenuation could be an artefact of the model, not a physical fact.

Next steps. (a) Request the raw helium SOBP survival tables from the Heidelberg group and, for each cell line/depth, fit two RSFs independently (one on K_{IDSB} , one on K_{cDSB}); test whether the K_{cDSB} -RSF is statistically zero. (b) Run isogenic pairs (DDRi-competent vs. DDRi-inhibited) at fixed dose across an LET ladder and fit the two-RSF variant. (c) If K_{cDSB} -RSF $\neq 0$, redo C4 with both parameters free.

Q2. *Is the peak of $RBE_{\text{DDRi}}/RBE_{\text{no-DDRi}}$ at $\approx 30 \text{ keV}/\mu\text{m}$ a physical prediction or a topological consequence of the giant-loop count?*

Basis. The GLOBLE geometry hard-codes ~ 3000 loops of 2 Mbp. The location of the RBE-ratio peak is sensitive to the crossover LET at which typical DSB counts per loop transition from 1 to ≥ 2 . That crossover depends on the loop count and size, which are themselves choices (Yokota/Sachs geometry).

Next steps. (a) Sweep loop count $\in \{1000, 3000, 6000, 12000\}$ and loop size $\in \{0.5, 1, 2, 4\}$ Mbp; report the peak LET as a function of both. (b) Compare against Hi-C-derived contact-domain sizes for the specific cell lines used (H460, H1437) — these lines have Hi-C data available. (c) Predict what peak LET *should* be seen if the loop geometry is replaced by cell-line-specific TAD structure, and design an experiment that would test it.

Q3. *What is the correct form of the RSF as a function of ATMi/DNA-PKi concentration — linear, saturating, or Hill-shaped?*

Basis. Liew 2019 Table 3 tabulates RSFs at 3 or 4 discrete inhibitor concentrations per line. UNIVERSE treats RSF as a per-condition input; it does not model the dose-response of the inhibitor itself. In clinical translation, physicians choose drug dose, not RSF. Without a dose $\rightarrow$ RSF map, C8 is unactionable.

Next steps. (a) Fit Hill and Michaelis–Menten forms to the H460 and H1437 ATMi RSF ladders and report goodness of fit; (b) validate/falsify with published PK/PD data on ATMi (e.g. AZD1390, AZD0156) — both agents have plasma concentration data; (c) extend UNIVERSE with a plasma.conc $\rightarrow$ RSF plugin and re-run C8 under realistic clinical drug schedules.

Q4. *Is the invariance of K_{cDSB} under DDRi consistent with the biology of NHEJ vs. MMEJ repair of clustered lesions, or is it a modelling convenience that breaks at high LET?*

Basis. The paper’s central mechanistic claim rests on K_{cDSB} being invariant under DDRi. Biologically, clustered DSBs may shift toward MMEJ (which uses Pol θ + ligase III, not the DNA-PKcs axis that ATMi/DNA-PKi target). If DDRi *does* affect clustered-lesion outcome via MMEJ competition, then the peak-then-fall shape of the RBE-ratio may be quantitatively wrong at high LET.

Next steps. (a) Repeat the H460 ATMi experiment in Pol θ -knockdown vs. wild-type at ^4He LET = 100 keV/ μm ; UNIVERSE predicts identical SF-ratios; MMEJ competition predicts divergent SF-ratios. (b) Extend UNIVERSE with a K_{cDSB} that depends on cluster complexity and repair-pathway availability, and refit the LET sweep. (c) Reconcile with Scholz 2020 LEM-IV predictions.

Q5. *Does the “DDRi + particle preserves therapeutic window” clinical claim (C8) survive when patient-scale organ-at-risk (OAR) heterogeneity, hypoxia, and inter-fraction repair are added to UNIVERSE?*

Basis. UNIVERSE as published treats a single homogeneous target voxel. Patient plans have LET spectra, OAR proximity, hypoxic sub-volumes (paper’s own Liew 2020 extension), and multi-fraction repair kinetics (Liew 2022). None of those enter C8’s recalculation as far as the abstract states. The clinical implication may not survive realistic patient-scale integration.

Next steps. (a) Reproduce the paper’s patient case with an open TPS surrogate (e.g. MatRad + FLUKA-compatible LET $_d$ maps) using a public phantom (e.g. TG-119); (b) add UNIVERSE-hypoxia (Liew 2020) as a voxel-level modifier; (c) add UNIVERSE-repair kinetics (Liew 2022) between fractions; (d) compare the DDRi + particle vs. DDRi + photon therapeutic-index numbers with vs. without each of those extensions. If the ordering flips under any extension, C8 is fragile.